DORA FRB HKMA PRA MAS

FINANCIAL RESILIENCE SCOPING ENGINE

Strategic Nexus

Global Regulated Enterprise (DORA, FRB, HKMA, PRA, MAS, APRA, NYDFS)
Threat-Led Penetration Testing Scoping Document — Nexus v33.0

ENTITY	RESILIENCE INDEX	ENGAGEMENT DURATION	CLASSIFICATION
Global Regulated Enterprise (DORA, FRB, HKMA, PRA, MAS, APRA, NYDFS)	100%	70 Days	Restricted

Financial Resilience Coverage Index

10

TLPT RESILIENCE INDEX

Financial Resilience Coverage Score

This score reflects the breadth of coverage across the four mandatory Threat-Led Penetration Testing pillars required under DORA Article 26, PRA CBEST, and HKMA iCAST.

TLPT COMPLIANCE PILLAR	COVERAGE STATUS
Threat Intelligence Profiling	VALIDATED
Covert Attack Emulation	VALIDATED
Purple Team Collaboration	VALIDATED
Executive Crisis Validation	VALIDATED

Executive Resourcing & Schedule Summary

The following summarizes the operational timeline and resourcing required to execute the defined Threat-Led Penetration Testing modules in alignment with regulatory mandates.

70
TOTAL ENGAGEMENT DAYS

14
EXECUTION WEEKS

4
TLPT PERSONNEL

4
ACTIVE MODULES

Automated TLPT Resourcing Allocation

ROLE	HEADCOUNT	TLPT RESPONSIBILITY	CLEARANCE
Threat Intel & Red Team Lead	1	Strategic oversight, regulatory coordination, TI report authorship, and final QA gate review.	Full Knowledge
Covert Red Team Operator	3	Technical execution, assumed breach operations, exploit development, and stealth persistence.	Full Knowledge
Purple Team Facilitator	1	Replay exercise facilitation, SOC detection tuning, and SIEM rule validation.	Full Knowledge
Client White Team	TBD	Kill-switch authority, CBS owner coordination, and regulatory notification.	Full Knowledge

1.5 Regulatory Alignment: Global Regulated Enterprise (DORA, FRB, HKMA, PRA, MAS, APRA, NYDFS)

Narrative: A unified global compliance scoping document consolidating DORA, FRB, HKMA, PRA, and MAS requirements.

REGULATORY BODY

Digital Operational Resilience Act (DORA) (European Union)

ARTICLE/REF	TITLE	REQUIREMENT	SCOPING GUIDANCE
Article 5	ICT Risk Management Framework	Firms must maintain a comprehensive, documented ICT risk management framework reviewed at least annually.	All critical ICT systems underpinning core financial functions.
Article 9	Protection & Prevention	Continuous monitoring and safeguarding of ICT systems; anomaly detection and prevention controls.	SIEM / SOC controls and detection capabilities under covert Red Team conditions.
Article 11	Business Continuity & Disaster Recovery	Firms must implement and test BCPs and DR plans for all critical functions.	RTO/RPO objectives tested via Purple Team replay and IR tabletop exercises.
Article 24	General ICT Testing Programme	Annual ICT resilience testing programme with risk-based scope.	All critical ICT systems and third-party ICT providers.
Article 26	Threat-Led Penetration Testing (TLPT)	Advanced TLPT required every 3 years for major financial entities. Mandates intelligence-led, covert, scenario-based Red Team testing.	Qualified external Red Team with regulatory oversight. TI-PHASE must precede RT-COVERT.
Article 28	ICT Third-Party Risk Management	Contractual obligations and exit strategies must be defined for all critical ICT third-party providers.	Cloud workloads, payment processors, and external ICT dependencies.
Article 30	Key Contractual Provisions	Contracts with ICT third parties must define SLAs, audit rights, and regulatory access provisions.	All contractual frameworks with critical ICT third-party service providers.

Mandatory Evidence Artifacts:

- Asset Register & CBS Map
- ICT Risk Register
- Threat Intelligence Profile
- TLPT Test Plan (Pre-Execution)
- TLPT Execution Report
- Remediation Plan & Timeline
- Third-Party ICT Risk Assessment

TIBER-EU Alignment & ESA Oversight

DORA Article 26 is underpinned by the TIBER-EU framework developed by the European Central Bank. Engagements must align with TIBER-EU's three-phase structure: Preparation (scoping & TI procurement), Test (covert Red Team execution), and Closure (reporting & remediation). The Joint Supervisory Team (JST) or relevant ESA (EBA/ESMA/EIOPA) must be notified prior to testing commencing on systemically important entities.

ICT Asset Criticality Mapping & CIF Declaration

Scoping must begin with a formally approved ICT Asset Register. Every asset in scope must be mapped to a Critical or Important Function (CIF) as defined under DORA Article 3. If an asset supports a CIF, it is subject to enhanced TLPT requirements and must appear in the CBS register. Assets that cannot be mapped to a CIF must be formally excluded with documented justification.

Third-Party ICT Service Provider Authorization

DORA Article 28 mandates that contractual arrangements with critical ICT third-party providers include explicit clauses permitting security testing. This scoping pack must include a Third-Party Dependency Register and written confirmation from each relevant provider. Cloud service providers hosting critical workloads must provide a Letter of Authorization (LoA) before any testing commences.

Threat Intelligence Sourcing Requirements

For DORA-compliant TLPT, threat intelligence must be sourced from a provider meeting the TIBER-EU TI Provider requirements. Intelligence must be current (within 12 months), specific to the financial sector, and targeted to the institution's geopolitical threat landscape. Generic CTI feeds are insufficient — bespoke Targeted Threat Intelligence Reports (TTIRs) are required.

Evidence & Artifact Requirements (Annex Submission)

Upon closure, the following must be submitted to the relevant ESA: (1) Final TLPT Report, (2) Remediation Plan with tracked findings, (3) Letter of Attestation from the TLPT Cyber Team (TCT), and (4) Management Summary for Board consumption. All artifacts must be retained for a minimum of 5 years under DORA Article 26(8).

REGULATORY BODY

Federal Reserve Board (FRB) Supervision (United States)

ARTICLE/REF	TITLE	REQUIREMENT	SCOPING GUIDANCE
SR 13-1	Supplemental Policy Statement on the Internal Audit Function	Independent testing and validation of technology risk controls. Audit function must be independent and objective.	Risk-based coverage of all material technology and cybersecurity controls.
SR 11-7	Guidance on Model Risk Management	Models used in risk decision-making must be subject to rigorous validation and independent testing.	AI/ML fraud detection models and algorithmic trading controls.
CAT	Cybersecurity Assessment Tool (CAT)	Institutions must assess their cybersecurity maturity against the FFIEC CAT framework.	Baseline controls, evolving controls, and innovative controls across all technology domains.

Mandatory Evidence Artifacts:

- Independent Technology Review Report
- Management Response & Remediation Plan
- Prior Findings Remediation Evidence
- FFIEC CAT Baseline Assessment

SR 13-1 Independence & Objectivity Requirements

Federal Reserve SR Letter 13-1 requires that security assessments are performed by individuals who are independent from the functions being tested and free from conflicts of interest. For institutions under FRB supervision, the testing provider must not have designed, implemented, or maintained the controls under assessment. Independence attestations must be documented and available for Federal Reserve examination.

CCAR & Supervisory Stress Test Alignment

For Large Bank Holding Companies (LBHCs) subject to the Comprehensive Capital Analysis and Review (CCAR), cybersecurity testing scope should align with the technology systems supporting capital planning, stress testing models, and reporting infrastructure. Any vulnerabilities identified in systems supporting CCAR submissions must be escalated to the Board Risk Committee and the Federal Reserve within defined timelines.

FFIEC Cybersecurity Assessment Tool (CAT) Integration

FRB-supervised institutions are expected to maintain an FFIEC CAT assessment aligned with their risk profile. The CAT maturity level must inform the depth of penetration testing — institutions at Baseline maturity should address foundational gaps; Intermediate and Advanced maturity institutions are expected to conduct more sophisticated adversarial simulation exercises. CAT results must be available to the FRB examiner upon request.

Technology Service Provider (TSP) Supervision

The FRB, in coordination with the FDIC and OCC through the FFIEC, supervises Technology Service Providers (TSPs) that support federally regulated institutions. For scoping purposes, all critical TSPs must be included in the Third-Party ICT register. The FRB may conduct independent examinations of TSPs and may require the production of penetration testing reports and remediation evidence from the TSP.

Management Response & Board Reporting

FRB examination expectations require that all identified security findings receive a formal management response with an estimated remediation timeline. Critical findings must be reported to the Board of Directors or its designated Risk Committee within 30 days of the final report issuance. Prior findings from FRB examinations that remain unresolved must be formally tracked and evidence of progress documented at each subsequent examination.

Hong Kong Monetary Authority (HKMA) iCAST (Hong Kong)

ARTICLE/REF	TITLE	REQUIREMENT	SCOPING GUIDANCE
CFI 2.0 – Pillar 1	Cybersecurity Assessment Framework (CAF)	Institutions must conduct a CAF self-assessment to determine their cybersecurity maturity level.	All critical systems and business functions serving retail and institutional clients.
CFI 2.0 – Pillar 2	Intelligence-led Cyber Attack Simulation (iCAST)	Perform controlled, covert, intelligence-led simulated attacks on live production systems.	Threat intelligence-derived simulation targeting CBS with systemic financial impact.
CFI 2.0 – Pillar 3	Cybersecurity Ecosystem Development	Contribution to the Hong Kong financial sector cybersecurity ecosystem and information sharing.	Participation in HKMA Cyber Intelligence Sharing Platform (CISP) post-engagement.

Mandatory Evidence Artifacts:

- CAF Self-Assessment Report
- Threat Intelligence Profile (TI-PHASE Output)
- iCAST Execution Log
- Closure Report & Control Effectiveness Summary
- Remediation Plan

CFI 2.0 Three-Pillar Architecture

The HKMA Cybersecurity Fortification Initiative 2.0 (CFI 2.0) is structured across three pillars: Pillar 1 — Cybersecurity Assessment Framework (CAF) self-assessment; Pillar 2 — Intelligence-led Cyber Attack Simulation Testing (iCAST); and Pillar 3 — Cybersecurity Ecosystem Development including participation in the HKMA Cyber Intelligence Sharing Platform (CISP). All three pillars are expected to be addressed in a complete iCAST engagement.

Assume Breach & SOC Detection Validation

iCAST mandates the 'Assume Breach' methodology to assess the detection and response capabilities of the institution's Security Operations Centre (SOC). The Red Team must simulate an adversary who has already gained initial access and must attempt lateral movement to CBS targets, with the Blue Team's detection speed and accuracy formally measured and documented in the execution log.

CAF Self-Assessment Prerequisite

Before an iCAST engagement, the institution must complete the HKMA CAF self-assessment to determine its cybersecurity maturity level. The CAF assessment score informs the depth and focus of the iCAST scope — institutions scoring in lower maturity bands must address foundational control gaps before a full iCAST can be conducted. CAF assessment evidence must be made available to the iCAST provider at scoping kickoff.

CISP Participation Post-Engagement

Following completion of the iCAST engagement, institutions are expected to contribute anonymized threat intelligence and findings to the HKMA Cyber Intelligence Sharing Platform (CISP) in accordance with Pillar 3 obligations. Threat actor TTPs, attack vectors, and detection gaps identified during the engagement should be formatted for CISP submission to benefit the broader Hong Kong financial ecosystem.

iCAST Notification & Regulatory Reporting

Institutions must notify the HKMA before commencing an iCAST engagement. Upon discovery of any critical or systemic vulnerability during execution, the institution must notify HKMA within 2 hours. Post-engagement, a formal report must be submitted to the HKMA within 30 days of the closure meeting, including the execution log, identified gaps, and a remediation timeline.

ARTICLE/REF	TITLE	REQUIREMENT	SCOPING GUIDANCE
CBEST – Phase 1	Threat Intelligence Assessment	Commission an independent Threat Intelligence assessment from a CREST-certified provider to define the threat landscape.	Identification of Threat Actor profiles, TTPs, and targeted Critical Business Services (CBS).
CBEST – Phase 2	Penetration Testing	Execute a covert, intelligence-led penetration test against the CBS identified in Phase 1.	CREST-certified Red Team with FCA-registered status. Zero knowledge for Blue Team.
CBEST – Phase 3	Remediation & Closure	Produce a Closure Report documenting findings, remediation actions, and control effectiveness.	Submitted to PRA/FCA within agreed timelines. Regulator retains right of review.
STAR-FS	Systematic Technical Assurance Review	A broader assessment framework for FMIs incorporating CBEST methodology with systemic risk focus.	Financial Market Infrastructures (FMIs) including payment systems and CCPs.

Mandatory Evidence Artifacts:

- Targeted Threat Intelligence Report (TTIR)
- CBEST Test Plan
- Red Team Execution Report
- Control Effectiveness Summary
- Closure Report (PRA/FCA Submission)

CREST Certification Mandate

All CBEST service providers — both the Threat Intelligence provider and the Penetration Testing provider — must hold current CREST certification relevant to their role. The Red Team provider must hold CREST Certified Simulated Attack Manager (CCSAM) or equivalent. Engagements conducted by uncertified providers are not recognized by the PRA or FCA as CBEST-compliant.

Bespoke Targeted Threat Intelligence Report (TTIR)

Phase 1 of CBEST requires commissioning a bespoke Targeted Threat Intelligence Report (TTIR) from an independent CREST-certified TI provider. The TTIR must identify specific threat actors targeting the institution, their TTPs, and map them to the institution's Critical Business Services. The TTIR is presented to and approved by the PRA/FCA before Red Team execution commences.

PRA/FCA Joint Regulatory Oversight

CBEST is subject to dual regulatory oversight from both the Prudential Regulation Authority (PRA) and the Financial Conduct Authority (FCA). Regulated firms must notify both authorities before commencing a CBEST assessment. The CBEST Testing Team (CTT) must maintain a formal communications channel with both regulators throughout the engagement, including pre-engagement briefing and post-closure debriefing.

STAR-FS for Financial Market Infrastructures

Financial Market Infrastructures (FMIs) — including central counterparties (CCPs), payment systems, and central securities depositories — are assessed under the STAR-FS (Systemic Testing of Assurance and Resilience for Financial Systems) framework, which applies CBEST methodology with a systemic risk lens. STAR-FS assessments require coordination with the Bank of England in addition to PRA/FCA oversight.

Closure Report & Remediation Tracking

The CBEST Closure Report must formally document all findings, the control effectiveness summary, and the prioritized remediation plan. Findings must be tracked to closure and evidence of remediation submitted to the PRA within agreed timelines. The firm must attest that critical-rated findings have been resolved or have a Board-approved risk acceptance in place before the next CBEST cycle.

ARTICLE/REF	TITLE	REQUIREMENT	SCOPING GUIDANCE
TRM Section 9	Cyber Risk Management	Institutions must establish a robust cyber risk management framework with Board-level oversight.	Board-level reporting on cyber risk and TLPT results mandatory.
TRM Section 12	Vulnerability Assessment and Penetration Testing (VAPT)	Conduct comprehensive VAPT on all critical IT systems annually or upon major system changes.	All internet-facing applications, core banking platforms, and payment APIs.
TRM Section 13	Cyber Incident Response & Management	Simulate adversary attacks to test the detection, containment, and recovery capabilities of the institution.	Red team exercises for major institutions. Incident response plan must be tested at least annually.
Notice MAS 655	Cyber Hygiene Baseline	Mandatory baseline cyber hygiene measures including network segmentation, patching, MFA, and logging.	Technical controls audit and validation as part of Red Team execution scope.

Mandatory Evidence Artifacts:

VAPT Report | Red Team Execution Log | Root Cause Analysis Report | Remediation Status Register | Board Cyber Risk Report

MAS Notice MAS 655 Baseline Cyber Hygiene Controls

MAS Notice MAS 655, effective 6 August 2020, mandates baseline cyber hygiene measures for all MAS-regulated financial institutions. These include: (1) network segmentation, (2) timely software patching, (3) multi-factor authentication on privileged and internet-facing systems, (4) malware protection, and (5) activity logging. VAPT scope must validate the effective implementation of all MAS 655 baseline controls.

TRM Section 12 — Vulnerability Assessment & Penetration Testing

MAS TRM Section 12 requires comprehensive VAPT on all critical IT systems, to be conducted at least annually or upon any major system change. VAPT must be conducted by qualified independent assessors. Results, including identified vulnerabilities and remediation actions, must be reported to senior management. Systems deemed critical under MAS definitions must be retested to verify remediation effectiveness.

Red Team Exercise Frequency & Scope

MAS TRM Section 13 requires major financial institutions to conduct Red Team exercises to test their detection, containment, and recovery capabilities. For institutions designated as major or systemically important, Red Team exercises should be conducted at least every 2 years. The scope must encompass internet-facing systems, core banking platforms, payment APIs, and data centres. Findings must be reviewed by the Board Risk Committee.

PDPA Compliance & Data Handling

All assessment data, execution logs, and findings reports must be handled in full compliance with Singapore's Personal Data Protection Act (PDPA). Any personal data of bank customers that is encountered during testing must be immediately reported to the White Team and excluded from further analysis. The engagement must include a formal Data Handling Protocol as an annex to this Scoping Document.

MAS Technology Risk Notification Obligations

Under MAS Notice MAS 644, institutions must notify MAS of any significant technology risk incidents within 1 hour of discovery and provide a root cause report within 14 days. Any critical vulnerability discovered during the TLPT engagement that poses an immediate systemic risk must be treated as a notifiable incident. The Red Team Lead must have a direct communication channel to the White Team for immediate escalation.

IMPLEMENTATION GUIDANCE

MITRE

ATT&CK Financial Sector Threat Groups

Execution scenarios must reference specific MITRE ATT&CK threat groups active against the financial sector. Key groups include: FIN7 (T1566, T1078, T1071), Carbanak (T1059, T1105), Lazarus Group (T1190, T1133, T1486), and TA505 (T1566.001, T1204). Attack chains must be constructed using the ATT&CK Navigator to demonstrate coverage across Initial Access, Execution, Persistence, Privilege Escalation, and Exfiltration tactics.

Kill Chain Mapping & TTP Evidence

Each attack scenario must produce a formal Kill Chain map documenting the exact sequence of ATT&CK techniques employed. Evidence of each technique must be captured in the form of tool output, command history, and screenshots. The Kill Chain document forms a mandatory annex to the TLPT Final Report and must be provided to the regulator upon request.

IMPLEMENTATION GUIDANCE

OWASP

A01:2021 — Broken Access Control

Testing must validate that users cannot act outside of their intended permissions. For financial institutions, this specifically includes testing of role-based access controls on banking APIs, separation of duties in payment authorization workflows, and privilege escalation paths on core banking administration interfaces.

A02:2021 — Cryptographic Failures

Testing must verify the protection of sensitive data in transit and at rest. For regulated financial data, this includes PAN data (PCI-DSS scope), customer PII, and interbank settlement data. Weak cipher suites, expired TLS certificates, and improperly stored credentials must all be tested.

A10:2021 — Server-Side Request Forgery (SSRF)

SSRF vulnerabilities are particularly dangerous in financial API environments as they can allow an attacker to pivot from a public-facing API to internal payment infrastructure. All public-facing Open Banking APIs must be tested for SSRF to prevent lateral movement from external to internal payment networks.

IMPLEMENTATION GUIDANCE

CIS

Control 1 & 2: Asset & Software Inventory

Actively manage all enterprise assets and software. For TLPT engagements, the CBS Register serves as the asset inventory baseline. Any asset discovered during the Red Team execution that was not in the CBS Register must be flagged as a shadow IT finding and reported as a governance gap.

Control 6: Access Control Management

Testing must validate privileged access management (PAM) controls, MFA enforcement on all privileged accounts, and the absence of default or shared credentials on network devices and servers.

Control 17: Incident Response Management

The Purple Team replay exercises must test the institution's incident response capabilities against the CIS Control 17 framework, measuring time-to-detect, time-to-contain, and time-to-recover against the defined TLPT attack scenarios.

DORA Implementation Guidance

TIBER-EU Alignment & ESA Oversight

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Technology Service Provider (TSP) Supervision

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HKMA Implementation Guidance

CFI 2.0 Three-Pillar Architecture

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iCAST mandates the 'Assume Breach' methodology to assess the detection and response capabilities of the institution's Security Operations Centre (SOC). The Red Team must simulate an adversary who has already gained initial access and must attempt lateral movement to CBS targets, with the Blue Team's detection speed and accuracy formally measured and documented in the execution log.

CAF Self-Assessment Prerequisite

Before an iCAST engagement, the institution must complete the HKMA CAF self-assessment to determine its cybersecurity maturity level. The CAF assessment score informs the depth and focus of the iCAST scope — institutions scoring in lower maturity bands must address foundational control gaps before a full iCAST can be conducted. CAF assessment evidence must be made available to the iCAST provider at scoping kickoff.

CISP Participation Post-Engagement

Following completion of the iCAST engagement, institutions are expected to contribute anonymized threat intelligence and findings to the HKMA Cyber Intelligence Sharing Platform (CISP) in accordance with Pillar 3 obligations. Threat actor TTPs, attack vectors, and detection gaps identified during the engagement should be formatted for CISP submission to benefit the broader Hong Kong financial ecosystem.

iCAST Notification & Regulatory Reporting

Institutions must notify the HKMA before commencing an iCAST engagement. Upon discovery of any critical or systemic vulnerability during execution, the institution must notify HKMA within 2 hours. Post-engagement, a formal report must be submitted to the HKMA within 30 days of the closure meeting, including the execution log, identified gaps, and a remediation timeline.

PRA Implementation Guidance

CREST Certification Mandate

All CBEST service providers — both the Threat Intelligence provider and the Penetration Testing provider — must hold current CREST certification relevant to their role. The Red Team provider must hold CREST Certified Simulated Attack Manager (CCSAM) or equivalent. Engagements conducted by uncertified providers are not recognized by the PRA or FCA as CBEST-compliant.

Bespoke Targeted Threat Intelligence Report (TTIR)

Phase 1 of CBEST requires commissioning a bespoke Targeted Threat Intelligence Report (TTIR) from an independent CREST-certified TI provider. The TTIR must identify specific threat actors targeting the institution, their TTPs, and map them to the institution's Critical Business Services. The TTIR is presented to and approved by the PRA/FCA before Red Team execution commences.

PRA/FCA Joint Regulatory Oversight

CBEST is subject to dual regulatory oversight from both the Prudential Regulation Authority (PRA) and the Financial Conduct Authority (FCA). Regulated firms must notify both authorities before commencing a CBEST assessment. The CBEST Testing Team (CTT) must maintain a formal communications channel with both regulators throughout the engagement, including pre-engagement briefing and post-closure debriefing.

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Financial Market Infrastructures (FMIs) — including central counterparties (CCPs), payment systems, and central securities depositories — are assessed under the STAR-FS (Systemic Testing of Assurance and Resilience for Financial Systems) framework, which applies CBEST methodology with a systemic risk lens. STAR-FS assessments require coordination with the Bank of England in addition to PRA/FCA oversight.

Closure Report & Remediation Tracking

The CBEST Closure Report must formally document all findings, the control effectiveness summary, and the prioritized remediation plan. Findings must be tracked to closure and evidence of remediation submitted to the PRA within agreed timelines. The firm must attest that critical-rated findings have been resolved or have a Board-approved risk acceptance in place before the next CBEST cycle.

MAS Implementation Guidance

MAS Notice MAS 655 Baseline Cyber Hygiene Controls

MAS Notice MAS 655, effective 6 August 2020, mandates baseline cyber hygiene measures for all MAS-regulated financial institutions. These include: (1) network segmentation, (2) timely software patching, (3) multi-factor authentication on privileged and internet-facing systems, (4) malware protection, and (5) activity logging. VAPT scope must validate the effective implementation of all MAS 655 baseline controls.

TRM Section 12 — Vulnerability Assessment & Penetration Testing

MAS TRM Section 12 requires comprehensive VAPT on all critical IT systems, to be conducted at least annually or upon any major system change. VAPT must be conducted by qualified independent assessors. Results, including identified vulnerabilities and remediation actions, must be reported to senior management. Systems deemed critical under MAS definitions must be retested to verify remediation effectiveness.

Red Team Exercise Frequency & Scope

MAS TRM Section 13 requires major financial institutions to conduct Red Team exercises to test their detection, containment, and recovery capabilities. For institutions designated as major or systemically important, Red Team exercises should be conducted at least every 2 years. The scope must encompass internet-facing systems, core banking platforms, payment APIs, and data centres. Findings must be reviewed by the Board Risk Committee.

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MAS Technology Risk Notification Obligations

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MITRE Implementation Guidance

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Kill Chain Mapping & TTP Evidence

Each attack scenario must produce a formal Kill Chain map documenting the exact sequence of ATT&CK techniques employed. Evidence of each technique must be captured in the form of tool output, command history, and screenshots. The Kill Chain document forms a mandatory annex to the TLPT Final Report and must be provided to the regulator upon request.

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A01:2021 — Broken Access Control

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SSRF vulnerabilities are particularly dangerous in financial API environments as they can allow an attacker to pivot from a public-facing API to internal payment infrastructure. All public-facing Open Banking APIs must be tested for SSRF to prevent lateral movement from external to internal payment networks.

CIS Implementation Guidance

Control 1 & 2: Asset & Software Inventory

Actively manage all enterprise assets and software. For TLPT engagements, the CBS Register serves as the asset inventory baseline. Any asset discovered during the Red Team execution that was not in the CBS Register must be flagged as a shadow IT finding and reported as a governance gap.

Control 6: Access Control Management

Testing must validate privileged access management (PAM) controls, MFA enforcement on all privileged accounts, and the absence of default or shared credentials on network devices and servers.

Control 17: Incident Response Management

The Purple Team replay exercises must test the institution's incident response capabilities against the CIS Control 17 framework, measuring time-to-detect, time-to-contain, and time-to-recover against the defined TLPT attack scenarios.

Threat Scenarios & Modeling

PART I.C: THREAT INTELLIGENCE

Threat Intelligence Profiles

> **MANDATORY REQUIREMENT PENDING:** A Threat Intelligence phase or report must be completed and formally appended to define the attack scenarios before Red Team or Intelligence-Led testing begins.

2.1 Digital Chain of Evidence

Artifacts produced during this engagement follow a cryptographically-verifiable lifecycle.

LIFECYCLE PHASE	VALIDATION METHOD
Payload & Exploit Transmission	Zero-Knowledge End-to-End Encryption (E2EE)
Exfiltrated Data Storage	Ephemeral Air-Gapped Analysis Environment (256-bit AES)
Red Team Telemetry Retention	Secure SOC Aggregation (Strict 30-Day Limit)
Artifact Destruction	NIST 800-88 Compliant Cryptographic Wipe

2.2 Data Sovereignty & Residency

To adhere to regulatory data protection requirements, the following residency rules apply to all engagement artifacts.

FRAMEWORK	RESIDENCY MANDATE
DORA	All TLPT evidence and reporting artifacts must remain strictly within EU boundaries or approved adequate jurisdictions. Cloud storage requires EU-based tenants under GDPR compatibility.
FRB	Federal regulatory artifacts and all supporting working papers must be accessible on-shore within the United States and subject to Federal Reserve examination.
HKMA	Assessment data and execution artifacts should be stored locally in Hong Kong in accordance with HKMA guidelines and the Personal Data (Privacy) Ordinance (PDPO).
PRA	All testing artifacts, raw execution logs, and threat intelligence reports must remain strictly within the United Kingdom. Cross-border data transfer requires explicit PRA approval.
MAS	Financial data and all assessment details must comply with the Singapore PDPA and MAS Outsourcing Guidelines. On-shore processing is strongly preferred; offshore processing requires MAS notification.

2.3 Regulator Audit & Access Rights

The following audit rights and access privileges are contractually granted to the respective regulatory bodies.

REGULATOR	ACCESS RIGHTS
DORA	European Supervisory Authorities (ESAs) — EBA, ESMA, and EIOPA — must have full, unrestricted, and on-demand access to all TLPT results, threat intelligence profiles, and remediation evidence.
FRB	Federal Reserve examiners possess unrestricted rights to review all working papers, final security deliverables, and management responses regardless of third-party engagement.
HKMA	HKMA retains the right to directly review threat intelligence models, execution logs, and remediation plans. Institutions must notify HKMA of any critical vulnerabilities discovered during iCAST.
PRA	PRA and FCA maintain joint audit rights over the testing methodology, control mappings, and all CBEST deliverables. CREST certification of the service provider is a regulatory prerequisite.
MAS	MAS requires immediate notification of any critical vulnerabilities and retains the right to independently audit all VAPT reports, Red Team logs, and remediation evidence.

Operational Team Governance

PART II.A: TEAM STRUCTURE

White Team & Control Group Structure

In accordance with intelligence-led frameworks (CBEST/iCAST), the engagement must be overseen by a strictly segregated Control Group to ensure operational safety without alerting the defensive Blue Team.

GOVERNANCE BODY	REGULATORY RESPONSIBILITY	CLEARANCE LEVEL
The White Team	Engagement sponsors holding the emergency Kill-Switch. Responsible for defining the Critical Business Services (CBS) and Impact Tolerances.	Full Knowledge
The Control Group	A trusted subset of the client's internal IT/Security team who monitor the engagement for safety but do not participate in defense.	Full Knowledge
The Blue Team	The operational SOC and incident responders who must defend the network as if under a real attack.	Zero Knowledge
Threat Intel Provider	Independent vendor supplying the threat landscape profiles and attack scenarios to the Red Team.	Execution Phase
Covert Red Team	The execution unit simulating the Threat Actor's TTPs against the CBS.	Execution Phase

Critical Business Services & Impact Tolerance

PART II.C: CBS IMPACT TOLERANCE

Critical Business Services (CBS) Scope

⚠ **ACTION REQUIRED — CBS DECLARATION MISSING:** No Critical Business Services have been declared. Under DORA Article 26 and PRA CBEST, the client must map target infrastructure to its underlying business function and formally define the maximum impact tolerance before testing may begin.

Access & Test Data Readiness

> Standard testing prerequisites apply: Client must provide stable network access (VPN/Jump host), valid application credentials (min. 2 per role), and any required API documentation prior to the start of the testing window.

Third-Party ICT Supply Chain Scope

PART II.D: THIRD-PARTY RISK

Third-Party ICT Dependency Register

> No third-party ICT dependencies or hosted environments have been declared in scope. Testing is restricted to first-party infrastructure.

Technical Scope of Work

TI-PHASE — **Targeted Threat Intelligence Gathering**

Formally scoped intelligence gathering phase designed to identify threat actors, define attack scenarios, and produce the Threat Intelligence Profile required before Red Team execution.

Allocated Schedule: 15 Days

RT-COVERT — **Covert Red Team Execution**

Intelligence-led, objective-based penetration testing executing the scenarios defined in the TI-PHASE. Includes stealth, evasion, and assumed-breach scenarios against core financial systems.

Allocated Schedule: 40 Days

PT-REPLAY — **Purple Team Replay Exercises**

Collaborative replay of Red Team attack chains alongside the defensive Blue Team to validate SOC detections, incident response capabilities, and SIEM rule effectiveness.

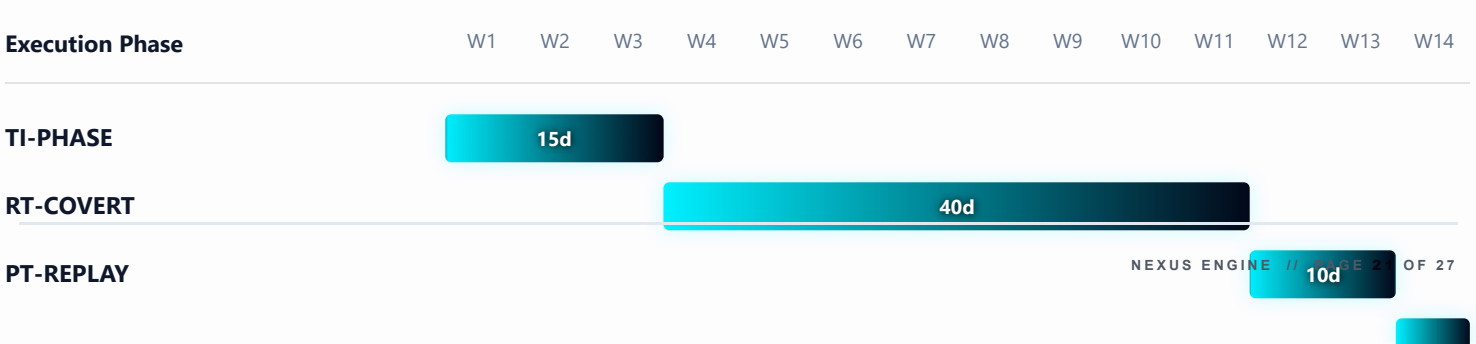
Allocated Schedule: 10 Days

IR-VALIDATION — **Incident Response & Crisis Validation**

Executive tabletop and crisis management simulation based on the findings from the Red Team execution to validate Board-level incident response maturity.

Allocated Schedule: 5 Days

Execution Timeline (Gantt)



Financial Control Mapping (Omni Matrix)

PART III.B: OMNI MATRIX

Financial Control Mapping (Omni Matrix)

The following TLPT-specific financial controls are in scope for the selected execution modules, mapped to their regulatory article references.

CONTROL ID	CONTROL NAME	FRAMEWORK	REGULATORY REFERENCE	IMPACT RATING
CB_01	Core Banking Network Segmentation	DORA	Art. 24 (TLPT)	**Critical**
CB_02	Retail Payment Infrastructure Isolation	DORA	Art. 26 (ICT Resilience)	**Critical**
SW_01	SWIFT Gateway Network Isolation	PRA CBEST	Objective A – Segregation	**Critical**
SW_02	SWIFT Messaging Integrity Validation	PRA CBEST	Objective B – Data Integrity	**Critical**
IAM_01	Privileged Access Management (PAM)	FRB	SR 13-1 – Access Control	**High**
IAM_02	Service Account Credential Hygiene	MAS TRM	Section 12.3 – IAM	**High**
IAM_03	Multi-Factor Authentication Enforcement	MAS TRM	Section 12.4 – Authentication	**High**
TI_01	Threat Intelligence Profile Development	HKMA iCAST	CFI 2.0 – TI Framework	**High**
TI_02	Threat Actor Attribution & Profiling	PRA CBEST	CBEST TI Module	**High**
TI_03	Attack Scenario Development (MITRE ATT&CK)	HKMA iCAST	CFI 2.0 – Scenario Modelling	**High**
TI_04	Threat-Led Defense Posture Assessment	HKMA iCAST	CFI-2 – Posture Review	**High**
IR_01	Active Incident Response Capabilities	DORA	Art. 26 – Crisis Management	**Medium**
IR_02	Executive Crisis Management Simulation	PRA CBEST	CBEST – Closure Report	**Critical**
IR_03	SOC Detection & SIEM Validation	MAS TRM	Section 13 – Incident Response	**High**
IR_04	Purple Team Replay & Remediation Verification	FRB	SR 11-7 – Model Validation	**Medium**
DR_01	Disaster Recovery Plan Validation	DORA	Art. 11 – Business Continuity	**Critical**
DR_02	RTO / RPO Objective Testing Under Load	FRB	SR 13-1 – Continuity	**High**
API_01	Open Banking API Security Boundary	MAS TRM	Section 12.1 – API Controls	**High**
CLD_01	Cloud Workload Lateral Movement Detection	DORA	Art. 30 – ICT Third-Party Risk	**High**

RACI Responsibility Assignment Matrix

Defines who is Responsible, Accountable, Consulted, and Informed for each TLPT activity across all parties.

ACTIVITY	THREAT INTEL PROVIDER	COVERT RED TEAM LEAD	CLIENT WHITE TEAM	CLIENT CONTROL GROUP	CLIENT SECURITY OPS (SOC)	CLIENT EXEC SPONSOR / CISO
Targeted Threat Intelligence Gathering	R	C	A	I	I	I
Attack Scenario & TTP Development	R	C	A	I	I	I
Critical Business Service (CBS) Declaration	I	I	A	C	C	R
Regulator & Scope Notification	I	C	A	I	I	R
White Team Briefing & Kill-Switch Assignment	I	C	R	C	I	A
Covert Red Team Execution (TLPT)	C	R	A	I	I	I
Emergency Kill-Switch Activation	I	C	R	C	I	A
Purple Team Replay Exercises	C	R	I	I	A	I
SOC Detection Rate Validation	I	C	A	C	R	I
Executive Crisis Management Tabletop	I	C	A	I	I	R
Remediation Planning & Sign-off	I	C	C	I	R	A
Regulator Artifact Submission	C	C	I	I	R	A

Regulatory Deliverables Schedule

The following reports and evidence logs are contractually required to satisfy the loaded regulatory frameworks.

REGULATORY BODY	MANDATORY DELIVERABLE ARTIFACT
Digital Operational Resilience Act (DORA)	Asset Register & CBS Map
Digital Operational Resilience Act (DORA)	ICT Risk Register
Digital Operational Resilience Act (DORA)	Threat Intelligence Profile
Digital Operational Resilience Act (DORA)	TLPT Test Plan (Pre-Execution)
Digital Operational Resilience Act (DORA)	TLPT Execution Report
Digital Operational Resilience Act (DORA)	Remediation Plan & Timeline
Digital Operational Resilience Act (DORA)	Third-Party ICT Risk Assessment
Federal Reserve Board (FRB) Supervision	Independent Technology Review Report
Federal Reserve Board (FRB) Supervision	Management Response & Remediation Plan
Federal Reserve Board (FRB) Supervision	Prior Findings Remediation Evidence
Federal Reserve Board (FRB) Supervision	FFIEC CAT Baseline Assessment
Hong Kong Monetary Authority (HKMA) iCAST	CAF Self-Assessment Report
Hong Kong Monetary Authority (HKMA) iCAST	Threat Intelligence Profile (TI-PHASE Output)
Hong Kong Monetary Authority (HKMA) iCAST	iCAST Execution Log
Hong Kong Monetary Authority (HKMA) iCAST	Closure Report & Control Effectiveness Summary
Hong Kong Monetary Authority (HKMA) iCAST	Remediation Plan
Prudential Regulation Authority (PRA) CBEST	Targeted Threat Intelligence Report (TTIR)
Prudential Regulation Authority (PRA) CBEST	CBEST Test Plan
Prudential Regulation Authority (PRA) CBEST	Red Team Execution Report
Prudential Regulation Authority (PRA) CBEST	Control Effectiveness Summary
Prudential Regulation Authority (PRA) CBEST	Closure Report (PRA/FCA Submission)
Monetary Authority of Singapore (MAS) TRM	VAPT Report
Monetary Authority of Singapore (MAS) TRM	Red Team Execution Log
Monetary Authority of Singapore (MAS) TRM	Root Cause Analysis Report
Monetary Authority of Singapore (MAS) TRM	Remediation Status Register
Monetary Authority of Singapore (MAS) TRM	Board Cyber Risk Report

Quality Assurance & Peer Review

Formal Quality Assurance Gate

Based on the selected regulatory frameworks and execution modules, the following QA routing is algorithmically enforced.

REQUIRED QA LEVEL: TLPT Qualified Red Team Reviewer

Mandatory Threat-Led Penetration Testing (TLPT) review gate. Evidence and kill-chain paths must be validated by a certified Red Team peer reviewer before report release.

Verification & Sign-off

PART V: AUTHORIZATION

NEXUS SEAL

Executive Sponsor Approval

Lead Strategic Provider

Engagement Sign Off & Authorization

By signing below, the parties confirm that this Nexus Scoping Document has been reviewed and is accepted as the binding scope of work for the defined Threat-Led Penetration Testing engagement. This document satisfies the engagement scoping requirements for the active regulatory frameworks: **DORA, PRA CBEST, HKMA iCAST, FRB, and MAS TRM**.

SIGNATORY ROLE	FULL NAME	SIGNATURE	DATE
Client Executive Sponsor / CISO			
Client White Team Lead			
Red Team Lead (Provider)			
Threat Intelligence Lead (Provider)			

✓ **Document Control:** This document is classified **RESTRICTED**. Distribution is limited to parties named in the White Team register. Regulatory bodies retain audit rights as defined in Part II of this document.